

Programs & Apps on Linux

Now if you're following the PDFs, you'd know we're using a **certain VERSION** of **LINUX**.

- **Ubuntu** is the one we're using right now.
 - For *those on different versions*, **this will cover**; Debian/Ubuntu

Different *versions*, have slight **differences in all ways**. For Installing programs **the differences are small**. (*The online VM won't work anymore, no internet access*)

In ALL cases; you install & update programs from a **LIBRARY** of sorts (*App Store/Play Store in a sense*), from the **TERMINAL**.

Ubuntu/Debian: These versions are the most *friendly to beginners*.

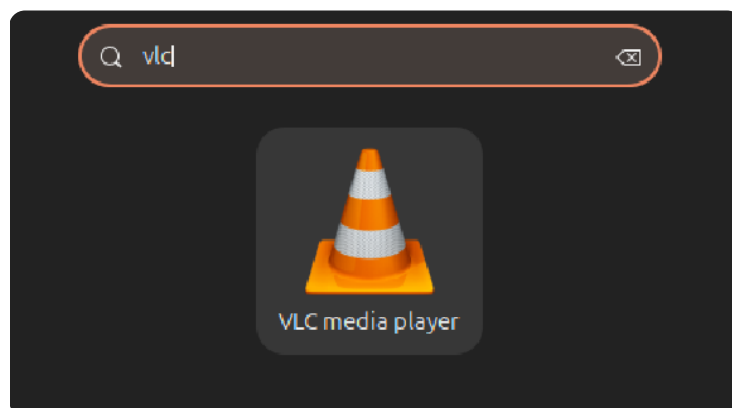
- The **LIBRARY** you get programs from, is called "**apt**" (*Advanced Package Tool*)
 - You see that "**package**" is **bolded**? That's because **LINUX** calls programs "**packages**".

Our First Command

> **sudo apt install "packagename" -y**

- **THIS MEANS**: As the owner of the **system**, for every user (**sudo**) from the **apt** library, **install** "**packagename**" and **DON'T** ask me yes or no (**-y**)

```
vboxuser@ExampleName:~$ sudo apt install vlc -y
```



BOOM! It's there.

- When trying to find the "**packagename**", look for them **online**; the apt search function is built for developers and is messy.

As you can see now, **it's installed**. Let's remove it for learning sake.

Our Second Command

> **sudo apt remove "packagename" -y** This REMOVES the installed package, no question asked.

```
vboxuser@ExampleName:~$ sudo apt remove vlc -y
[sudo] password for vboxuser:
```

Certain Packages *may NOT* exist in the APT LIBRARY.

```
vboxuser@ExampleName:~$ sudo apt install discord
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

No apt package "discord", but there is a snap with that name.
Try "snap install discord"

E: Unable to locate package discord
```

As *you can see ABOVE*, apt can *NOT find DISCORD*.

- But at the bottom, *highlighted that "snap" can see it?*

For **UBUNTU**, a *secondary LIBRARY*, called **snap** (pre-installed) *is used* for non-official programs.

- For **DEBIAN**, this must be installed manually, run: **sudo apt install snapd**

> **snap install discord** (*"remove" works as expected*)

```
vboxuser@ExampleName:~$ sudo snap install discord
discord 0.0.119 from Snapcrafters🌟 installed
```

- As you can see, *you don't need to type as many words as previous*. NO **SUDO** (*user only install, not every user, don't need to be the owner of the system*), and no questions (*no need for -y*)
-

Updating in **LINUX** updates for **ALL packages installed with that library**,

- It is a bit different for **WHICH library you used**.

For upgrading *packages installed with apt* run these two, to do so.

> **sudo apt update** Checks for any *updates*, **grabs the necessary packages** to install them from online.

> **sudo apt upgrade** Takes those packages from “update”, and actually runs the *upgrade* on **EVERYTHING**.

```
vboxuser@ExampleName:~$ sudo apt update
vboxuser@ExampleName:~$ sudo apt upgrade
```

For **snap**. This is a lot simpler.

> **snap refresh** Checks for any *updates*, and installs them in **ONE STEP**.

```
vboxuser@ExampleName:~$ snap refresh
```

That is **ALL you need** for **INSTALLING BASIC PACKAGES**.

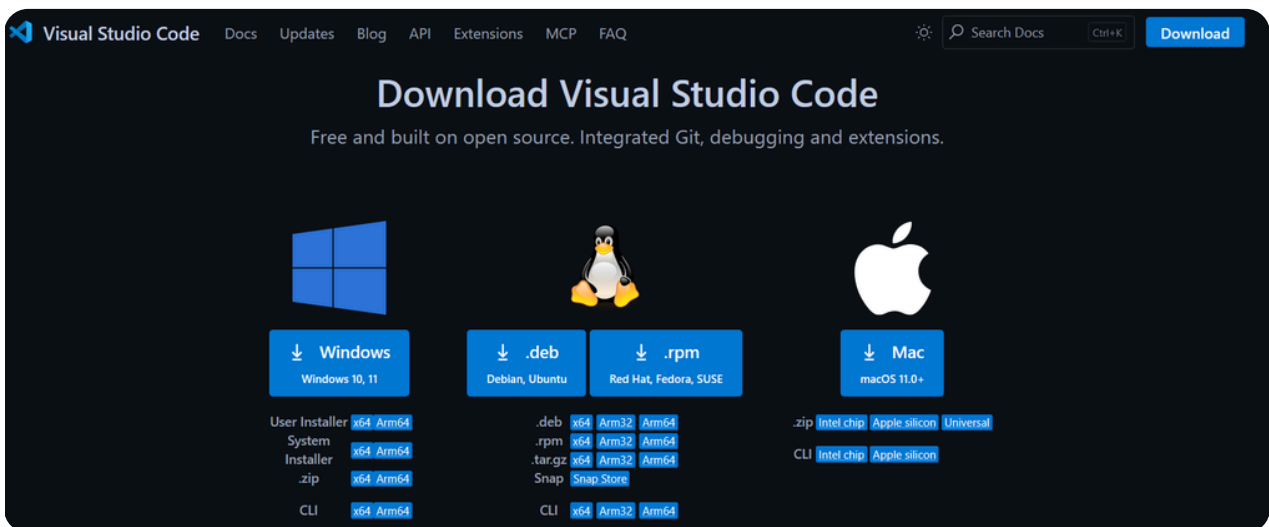
If you are **ONLY** using **UBUNTU/DEBIAN CASUALLY**, there is **NO NEED** for you to read the rest of this *PDF*. If you are a **security worried person** though, that is next.

If you're being picky, **snapt** isn't the USUAL way to install less popular packages on **DEBIAN**, due to how snap is a library managed by the public,
- So **ANYONE** can add programs.

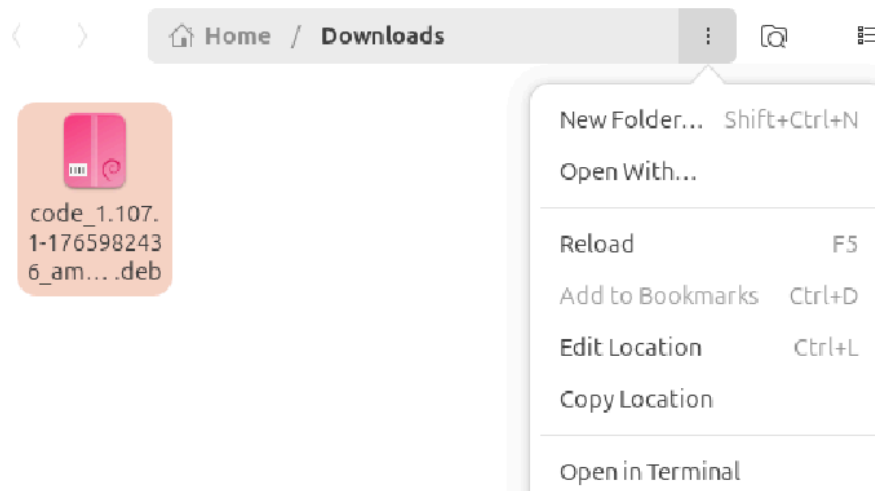
- Instead you use **.deb** files.
 - Found from the official Website of the APP you wish to download.
 - Get a **.deb** file, put it into your **DOWNLOADS** folder.
 - **OPEN THE TERMINAL**
 - Move to here **cd Downloads**
 - Run this with the **./FULL FILE NAME**, **sudo apt install ./FILENAME**

Example:

- **LOOK UP** the **VSCODE** installation page, Click on **.deb**

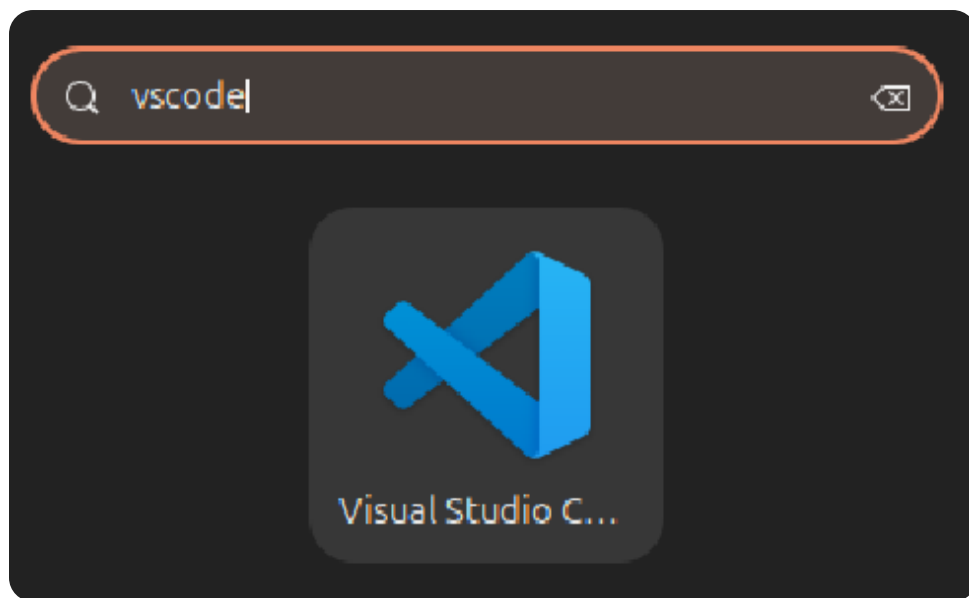


- **COPY** the file to get the file name. Open the **TERMINAL** from the **DOWNLOADS FOLDER**, your **.deb** should be there.



- Run **sudo apt install "FilePathYouCopied"** to **INSTALL** our package.

```
vboxuser@ExampleName:~/Downloads$ sudo apt install /home/vboxuser/Downloads/code_1.107.1-1765982436_amd64.deb
```



That is **EVERYTHING** you *really need to know about packages universally*,
SELECT programs *may have custom instructions*; but **by now** you have enough
knowledge to understand it.